

AMJAD ALOLAYAN

M.Sc. Sustainable Urban Development · GIS & Geospatial Analysis

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SUMMARY

Master's student in Sustainable Urban Development at Technical University Darmstadt with international experience in Japan, Vietnam, and Germany. Strong interest in GIS analysis, sustainable infrastructure systems, and life-cycle assessment (LCA), with growing expertise in Python and ArcGIS Pro for data-driven urban sustainability applications.

GIS PROJECTS

Multi-Criteria Solar Park Suitability Analysis – Wetteraukreis & Vogelsbergkreis, Hesse

2026

Self-directed portfolio project · ArcGIS Pro, Python (GeoPandas, Shapely, STRtree, Folium)

- Built a multi-criteria GIS suitability model for ground-mounted solar parks across two contrasting Hessian districts: six hard exclusions (water, settlements, forests, airports, protected areas via Hessen INSPIRE WFS, existing solar), parcel generation, and scoring of 197 candidates on five weighted criteria (EEG corridor, substation distance, slope, southness, irradiation) via zonal statistics and STRtree analysis in Python.
- Produced six cartographic ArcGIS Pro layouts and a live, interactive Folium/Leaflet web map; published the full analysis with documentation on GitHub and GitHub Pages.

Rooftop Solar Potential Analysis – Breuberg, Hesse

2026

Rebuilt and corrected university coursework as an independent portfolio project · ArcGIS Pro, PVGIS, OpenStreetMap

- Rebuilt a graded university solar-energy assignment from scratch in ArcGIS Pro, correcting a radiation-value error and re-deriving the formula chain using PVGIS 5.3 (PVGIS-SARAH3) 18-year average irradiation data; sourced OpenStreetMap building footprints via the Overpass API, reprojected to ETRS89/UTM Zone 32N, and calculated per-building solar PV potential through formula-driven attribute modelling.
- Designed cartographic layouts with corrected legends, methodology panel, and sources; identified that large industrial rooftops account for a disproportionate share of technical potential.

WORK EXPERIENCE

Compliance Officer, Gulf International Bank

Apr 2022 – Sep 2023

Al Khobar, Saudi Arabia

- Monitored client transactions using CRM systems and Oracle Financial Crime & Compliance (FCC); conducted policy and risk analysis, producing reports that supported regulatory compliance and internal governance.

Intern, Saudi Food and Drug Authority (SFDA)

May 2021 – Oct 2021

- Dammam, Saudi Arabia: supported regulatory projects on food and drug safety through data collection, policy documentation, and preparation of governmental reports.

Akiya 2.0

Sep 2018 – Feb 2019

- Hiroshima, Japan: contributed to the sustainable renovation of traditional buildings and assisted in preparing architectural as-built drawings using AutoCAD.

EDUCATION

M.Sc. Sustainable Urban Development – TU Darmstadt

Oct 2023 – Present (exp. 2027)

Darmstadt, Germany · Focus on sustainable urban/infrastructure development, climate mitigation, green building design, GIS, and Python-based environmental data analysis.

Joint Master's Program – Vietnamese-German University

Oct 2024 – Mar 2025

Vietnam · Sustainability, circular economy, decarbonisation strategies; applied GEG-based building energy calculations and cost-benefit analysis.

B.A. Community and Regional Policy Studies – Ritsumeikan University

Sep 2016 – Mar 2021

Osaka, Japan · Fully funded scholarship · Public policy, sustainability, and urban governance.

- Bachelor's thesis: "Adaptability of reverse-vending machines for sustainable waste management in Saudi Arabia," policy analysis and international case studies.

TECHNICAL SKILLS

GIS & Mapping: ArcGIS Pro (spatial analyst, zonal statistics, model building, cartographic layouts), QGIS, kernel density & multi-criteria suitability modelling

Programming: Python (GeoPandas, Shapely, STRtree, Folium, pandas, matplotlib), spatial data automation

Data, Tools & Workflows: Excel, SPSS, openLCA, OpenStreetMap/Overpass API, PVGIS, Git/GitHub, GitHub Pages; coordinate reprojection (ETRS89/UTM), spatial joins & nearest-neighbour analysis, raster/zonal statistics, web-map publishing (Leaflet)

LANGUAGES & CERTIFICATES

Arabic: Native · English: C1 · Japanese: C1 · German: A2–B1 (basic working proficiency)

GIS for Climate Action (Esri, 2026) · Geospatial Data in Python (DataCamp, 2026) · LCA Certificate (One Click LCA, 2026) · CSRD Fundamentals (2025)